CONDITION BASED ASSESSMENT
QUICK SCANNING REPORT
SBUM DIAGNOSTIC JOHOR
TNB BAHAGIAN PEMBAHAGIAN JOHOR

FUNCTIONAL LOCATION (ERMS)	:	JJBU/PCE/K00191
FUNCTIONAL LOCATION (SITE)	:	JJBU/PCE/K00191
SUBSTATION NAME (ERMS)	:	O/D KG MAJU JAYA NO.1
SUBSTATION NAME (SITE)	:	PE O/DOOR KG MAJU JAYA # 1
COORDINATE	:	1.583366, 103.718120
SUBSTATION TYPE	:	PCE
AREA	:	JOHOR BAHRU
STATE	:	JOHOR
PO	:	42234207
DATE	:	14 JUL 2026

SCANNED BY		
THERMOGRAPHY	:	MUHAMMAD OMAR RIDZUAN (2013MY20N010)

EQUIPMENT/TOOLS		
THERMOGRAPHY	:	FLIR T640 (SN 55912229) (CALIBRATION DUE DATE: 26 DEC 2026)

PREPARED BY		
NAME OF CONTRACTOR	:	**EET ENGINEERING SERVICES SDN BHD**
ADDRESS	:	NO.38, JALAN SS19/1D,
	:	47500, SUBANG JAYA, SELANGOR.

SUBSTATION SIGNBOARD



VISUAL DEFECT SUMMARY

NO.	EQUIPMENT	DEFECT DESCRIPTION	ELECTRICAL	NON-ELECTRICAL
1	RUBBER MAT	NO RUBBER MAT		X

SUBSTATION CONDITION

SUBSTATION OVERVIEW



SIGNBOARD



SWITCHGEAR



SWITCHGEAR NAMEPLATE



TRANSFORMER 1



TRANSFORMER 1 NAMEPLATE



SUBSTATION CONDITION

TRANSFORMER 2
N/A

TRANSFORMER 2 NAMEPLATE
N/A

FEEDER PILLAR TX1


FEEDER PILLAR TX1 NAMEPLATE
N/A

FEEDER PILLAR TX2
N/A

FEEDER PILLAR TX2 NAMEPLATE
N/A

SUBSTATION CONDITION

EFI	SF6 INDICATOR
	

BATTERY	BATTERY NAMEPLATE
N/A	N/A

FIRE EXTINGUISHER	FIRE EXTINGUISHER NAMEPLATE
	

SUBSTATION CONDITION

SWITCHGEAR PANEL NAME	LAST PREVENTIVE MAINTENANCE
	

VISUAL DEFECT

RUBBER MAT


NO RUBBER MAT

NORMAL/DEFECT STICKER

NORMAL STICKER


CBM CONDITION	81
	NORMAL

DISTRIBUTION SUBSTATION (PCE) CBA TEST SHEET

STATE / AREA:	JOHOR / JOHOR BAHRU	**DATE**	14/07/26			**FUNCTIONAL LOCATION:**
PE NAME:	O/D KG MAJU JAYA NO.1	**TIME IN / OUT**	2050	/	2140	JJBU/PCE/K00191
V.I REF. NO	81	**TEV BACK GROUND / HUMIDITY**	6 dB	/	68 %	**BACKGROUND TEMP : 32.9°C**

SWITCHGEAR

DETAIL								TEMPERATURE							ULTRASOUND			TEV			
NO.	**FDR. NO**	**FEEDER NAME**	**STATUS (TRIP / CLOSE)**	**LOAD (A)**	**CABLE TYPE/SIZE**	**HEATER (A)**	**SERIAL NO (ONLY BREAKER & OLU)**	**COMP**	**Tmin (°C)**	**Tmax (°C)**	**ΔT**	**Avg (°C)**	**FILE**	**REMARKS**	**dB**	**FILE**	**TYPE CHAR.**	**dB**	**PULSE/ CYCLE**	**TYPE CHAR.**	**REMARK**
1	2854	PDT KG.MAJU JAYA	CLOSE		XLPE			CABLE	31.3	31.4	0.1	31.4	248		0			6	0		
								BREAKER													
								TOP PANEL													
								PT													
2	2852	PAT KG MAJU JAYA NO.2	CLOSE		XLPE			CABLE	31.5	31.6	0.1	31.6	249		0			6	0		
								BREAKER													
								TOP PANEL													
								PT													
3	2853	TX 1000KVA	CLOSE		XLPE			CABLE	31.3	31.4	0.1	31.4	250		0			6	0		
								BREAKER													
								TOP PANEL													
								PT													
4								CABLE													
								BREAKER													
								TOP PANEL													
								PT													
OVERVIEW								CABLE				31.8	242								
								BREAKER													
								TOP PANEL				31.5	246								
								PT													

LOCAL TRANSFORMER

DETAILS					THERMOGRAHY					ULTRASOUND		DETAILS				THERMOGRAHY						ULTRASOUND		
TX ID	**SERIAL NO**	**CABLE TYPE/SIZE**	**REMARK**	**COMP.**	**Tmin(°C)**	**Tmax(°C)**	**ΔT**	**Avg (°C)**	**FILE**	**dB**	**TYPE CHAR.**	**TX ID**	**SERIAL NO**	**CABLE TYPE/SIZE**	**COMP.**	**Tmin(°C)**	**Tmax(°C)**	**ΔT**	**Avg (°C)**	**FILE**	**dB**	**FILE**	**TYPE CHAR.**	**REMARKS**
TX1	727812	XLPE		**HT CABLE**	32.0	32.2	0.2	32.1	254	0		**TX3**			**HT CABLE**									
				HT BUSHING	35.3	35.5	0.2	35.4	253	0					**HT BUSHING**									
		XLPE		**LV CABLE**	33.5	33.8	0.3	33.7	256						**LV CABLE**									
				LV BUSHING	35.8	36.1	0.3	36.0	255						**LV BUSHING**									
				BODY	37.5	42.7	5.2	40.1	251						**BODY**									
TX2				**HT CABLE**								**TX4**			**HT CABLE**									
				HT BUSHING											**HT BUSHING**									
				LV CABLE											**LV CABLE**									
				LV BUSHING											**LV BUSHING**									
				BODY											**BODY**									

LV DISTRIBUTION BOARD/FEEDER PILLAR/OTHERS

1	**LVDB / FP 1**	**INC. 1**	**INC. 2**	**INC. 3**	**F1**	**F2**	**F3**	**F4**	**F5**	**F6**	**F7**	**F8**	**F9**	**F10**
	CABLE TYPE / SIZE	XLPE			XLPE	XLPE	XLPE							
2	**LVDB / FP 2**	**INC. 1**	**INC. 2**	**INC. 3**	**F1**	**F2**	**F3**	**F4**	**F5**	**F6**	**F7**	**F8**	**F9**	**F10**
	CABLE TYPE / SIZE													

NO	**LVDB / FP**	**DEFECTIVE FEEDER NAME**	**Tmin(°C)**	**Tmax(°C)**	**ΔT**	**FILE**	**Avg (°C)**	**REMARK**
	INC 1							
	INC 2							
	F1							
	F2							
	F3							
	F4							
	F5							
	F6							
	F7							
	F8							
	Battery Bank							
	Black Box 1							
	Black Box 2							
	Lightning Arrester							
	Dropout Fuse							
	Auto Recloser							
	LBS							

FP TX1	**J-SLOTTED**
Manufacturer :	N/A
Serial No. :	N//A
Rating :	800A
LVDB TX2	
Manufacturer :	
Serial No. :	
Rating :	

Tested by (Name & Signature) :

EET ENGINEERING SERVICES SDN BHD
(534556-W)
No. 38-2, 2nd Floor, Jalan SS 19/1D
47500 Subang Jaya, Selangor.
Tel: 603-5636 6201 Fax: 603-5636 7205

TNB Supervisor/Engineer (Name & Signature) :

NOR AZMI BIN ABDUL RAHMAN
(N/P 10013600)
Pen. Juruteknik T/Kanan
Diagnostik (Substation - South)
Distribution Network Department, TNB

Ver. : 1-2019

Vi no mat,

TYPE PCE 81

VISUAL INSPECTION CHECKLIST FOR 11kV SUBSTATION

TENAGA NASIONAL BERHAD (200866-W)

DATE:	14/07/26												
P/E Name (ERMS):	O/D KG MAJU JAYA NO.1							F/L NO (ERMS):	JJBU/PCE/K00191				
P/E Name (SITE):	PE O/DOOR KG MAJU JAYA # 1							F/L NO (SITE):	JJBU/PCE/K00191				
GPS Coordinate:	1.583366, 103.718120												
PCE Type:	INDOOR			OUTDOOR	☑	COMPACT		ATTACH BUILDING		PAT		OTHER	
PCE Area	NORMAL	☑		INDUSTRIAL		COASTAL							
Switcgear Type 1:	RMU OIL			RMU SF6	☑	MRMU		VCB		OCB		OTHER	
Manufacturer/Model:	INDKOM			INS24									
Switchgear Manufactured Date:	03-08-2015					Switchgear Rating :		12kV				RMU SF6 / OIL S/N :	15072115279DE
Switcgear Type 2:	RMU OIL			RMU SF6		MRMU		VCB		OCB			
Manufacturer/Modal:													
Switchgear Manufactured Date:						Switchgear Rating :						RMU SF6 / OIL S/N :	
No. of Transformer:	1												
Tx 1:	Type	H/S		KVA	1000kVA	Construction Yr:	2015	Manufacturer:	SGB			S/N :	727812
Tx 2:	Type			KVA		Construction Yr:		Manufacturer:				S/N :	
Tx 3:	Type			KVA		Construction Yr:		Manufacturer:				S/N :	
Tx 4:	Type			KVA		Construction Yr:		Manufacturer:				S/N :	

KEJANGGALAN

1. Signboard:	Good	☑	Not Good		Remarks	
2. Earth connection:	Good	☑	Not Good		Remarks	
3. EFI Functioning:	Good	☑	Not Good		Remarks	
4. Heater Functioning:	Good		Not Good		Remarks	N/A
5. Labelling @Switchgear:	Good	☑	Not Good		Remarks	
6. Labelling @ LVDB:	Good	☑	Not Good		Remarks	
7. Gas Pressure:	Good	☑	Not Good		Remarks	
8. Tx Guard:	YES	☑	NO		Remarks	
9. Tx Oil Level (conservator tank):	Good		Not Good		Remarks	N/A
10. Tx Oil Leak:	YES		NO	☑	Remarks	
11. Tx Bushing:	Good	☑	Not Good		Remarks	
12. Top Demand Indicator (@LVDB):	Good	☑	Not Good		Remarks	
13. Link/fuse Holder:	Good	☑	Not Good		Remarks	
14. LV Termination:	Good	☑	Not Good		Remarks	
15. Rubber Mat:	YES		NO	☑	Remarks	
16. Trenching:	Good	☑	Not Good		Remarks	
17. Battery Voltage:	Good		Not Good		Remarks	N/A
18. Lighting:	Good	☑	Not Good		Remarks	
19. Roof Condition:	Good	☑	Not Good		Remarks	
20. Fire Fighting	Valid	☑	Expired		Remarks	Expiry Date: 11/9/26 No. of missing portable : -
21. Fence/door condition:	Good	☑	Not Good		Remarks	
22. Cleanliness/Bushes:	YES		NO	☑	Remarks	
23. Vandalisme	YES		NO	☑	Remarks	

24. Other Findings:

Tested by (Name & Signature) :

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(534556-W)
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47500 Subang Jaya, Selangor.
Tel: 603-5636 6200 Fax: 603-5636 7265

Date :

TNB Supervisor / Engineer (Name & Signature) :

NOR AZMI BIN ABDUL RAHMAN
(NP 10073600)
Sen. Juruteknik T/Kanan
Diagnostik (Substation - South)
Distribution Network Department, TNB

Date :

Ver. : 1-2019